

Workshop 8

F 9-10:50am

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Invalid Date

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In this workshop, we will answer the question: How do specific leaf area, water treatment, and species influence plant mass?

Specific leaf area is a continuous variable measured in cm^2/g .

Water treatment is a categorical variable (i.e. a *factor*) with 2 levels: drought stressed (DS) and well watered (WW).

Species is a categorical variable (again, a *factor*) with 7 levels.

Species name	Species code	Common name
<i>Encelia californica</i>	ENCCAL	Bush sunflower
<i>Eschsholzia californica</i>	ESCCAL	California poppy
<i>Penstemon centranthifolius</i>	PENCEN	Scarlet bugler
<i>Grindelia camporum</i>	GRINCAM	Gumweed
<i>Salvia leucophylla</i>	SALLEU	Purple sage
<i>Stipa pulchra</i>	STIPUL	Purple needlegrass
<i>Lotus scoparius</i>	LOTSCO	Deerweed

1. Set up

```
# read in your packages here
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(readxl) # reading excel files
library(scales) # modifying axis labels
library(ggeffects) # getting model predictions
library(MuMin) # model selection
```

```
# read in your data here
drought_exp <- read_xlsx(
  here("data",
    "Valliere_etal_EcoApps_Data.xlsx"),
  # use this to specify sheet within excel
  sheet = "First Harvest"
)
```

```
# saves it for all future plots
```

```
# storing colors to use for species
lotsco_col <- "#E69512"
pencen_col <- "#D6264F"
```

```

salleu_col <- "#6D397D"
enccal_col <- "#3A5565"
stipul_col <- "#3F564F"
esccal_col <- "#515481"
gricam_col <- "#6C91BD"

plant_colors <- c(
  "Lotus scoparius" = lotsco_col,
  "Penstemon centranthifolius" = pencen_col,
  "Salvia leucophylla" = salleu_col,
  "Encelia californica" = enccal_col,
  "Stipa pulchra" = stipul_col,
  "Eschscholzia californica" = esccal_col,
  "Grindelia camporum" = gricam_col
)

# storing colors to use for water treatments
ds_col <- "#A62F03"
ww_col <- "#045CB4"

# storing a ggplot theme (that will be used for all ggplots)
theme_set(theme_bw())

```

2. Clean data

Data source: Valliere, Justin; Zhang, Jacqueline; Sharifi, M.; Rundel, Philip (2019). Data from: Can we condition native plants to increase drought tolerance and improve restoration success? [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v0861f7>

```

# create new object called drought_exp_clean
drought_exp_clean <- drought_exp |>
  # lowercase letters in columns, replace spaces with underscores
  clean_names() |>
  # create new column with species full names
  mutate(species_name = recode_values(
    species,
    "ENCCAL" ~ "Encelia californica", # bush sunflower
    "ESCCAL" ~ "Eschscholzia californica", # California poppy
    "PENCEN" ~ "Penstemon centranthifolius", # Scarlet bugler
    "GRICAM" ~ "Grindelia camporum", # Gumweed
    "SALLEU" ~ "Salvia leucophylla", # purple sage
  ))

```

```

  "STIPUL" ~ "Stipa pulchra", # purple needlegrass
  "LOTSCO" ~ "Lotus scoparius" # deerweed
)) |>
# create new column with descriptive water treatment names
mutate(water_treatment = recode_values(
  water,
  "WW" ~ "Well watered",
  "DS" ~ "Drought stressed"
)) |>
# set levels for linear model using species as a factor,
# relevel allows us to choose levels within factor, reference is lotus scoparius
mutate(species_name = as_factor(species_name),
       species_name = fct_relevel(species_name,
                                   "Lotus scoparius",
                                   "Penstemon centranthifolius",
                                   "Salvia leucophylla",
                                   "Encelia californica",
                                   "Stipa pulchra",
                                   "Eschscholzia californica",
                                   "Grindelia camporum")) |>

# do same thing for water treatment
mutate(water_treatment = as_factor(water_treatment),
       water_treatment = fct_relevel(water_treatment,
                                      "Drought stressed",
                                      "Well watered")) |>

# keep relevant columns: species, water treatment, specific leaf area, plant mass
select(species_name, water_treatment, sla, total_g)

```

Insert code to double check the structure of the data frame here.

```
str(drought_exp_clean)
```

```

tibble [70 x 4] (S3: tbl_df/tbl/data.frame)
 $ species_name   : Factor w/ 7 levels "Lotus scoparius",...: 4 4 4 4 4 4 4 4 4 4 ...
 $ water_treatment: Factor w/ 2 levels "Drought stressed",...: 2 2 2 2 2 1 1 1 1 1 ...
 $ sla            : num [1:70] 170 215 209 216 222 ...
 $ total_g       : num [1:70] 0.455 0.329 0.45 0.359 0.352 ...

```

Factors and levels display properly, so we're set for the linear model. This is the only way to check.

Insert code to display 10 random rows from the data frame here.

```
# display 10 rows from data frame
slice_sample(
  drought_exp_clean,
  n = 10
)
```

```
# A tibble: 10 x 4
```

	species_name <fct>	water_treatment <fct>	sla <dbl>	total_g <dbl>
1	Penstemon centranthifolius	Well watered	160.	0.172
2	Stipa pulchra	Well watered	234.	0.545
3	Salvia leucophylla	Well watered	96.2	0.232
4	Penstemon centranthifolius	Drought stressed	146.	0.126
5	Encelia californica	Well watered	216.	0.359
6	Stipa pulchra	Well watered	159.	0.353
7	Encelia californica	Well watered	170.	0.455
8	Salvia leucophylla	Drought stressed	74.4	0.131
9	Lotus scoparius	Well watered	123.	0.182
10	Lotus scoparius	Well watered	106.	0.185

3. Visualizing data

What is the relationship between SLA and plant mass?

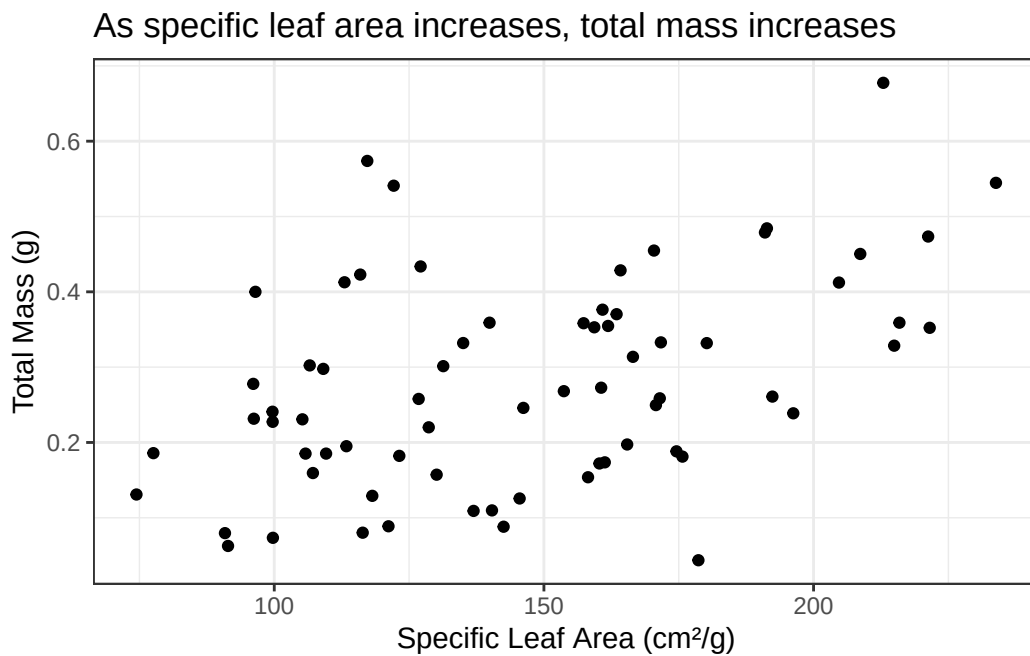
Make a scatterplot to visualize SLA (the predictor) on the x-axis, and total mass (the response) on the y-axis.

Label the axes with units where appropriate.

Add a title describing the relationship (e.g. as SLA increases, total mass _____).

```
# base layer for scatterplot
# relating specific leaf area (x) to total mass (y)
ggplot(
  data = drought_exp_clean,
  mapping = aes(
    x = sla,
    y = total_g,
  )
) +
  # add points
```

```
geom_point() +
# rename axes, add title, the \U00B2/ is unicode for a superscript
labs(
  x = "Specific Leaf Area (cm\U00B2/g)",
  y = "Total Mass (g)",
  title = "As specific leaf area increases, total mass increases"
)
```



What are the differences in total mass between water treatments?

Make a jitterplot to visualize the differences in plant mass between water treatments.

Label the axes with units where appropriate.

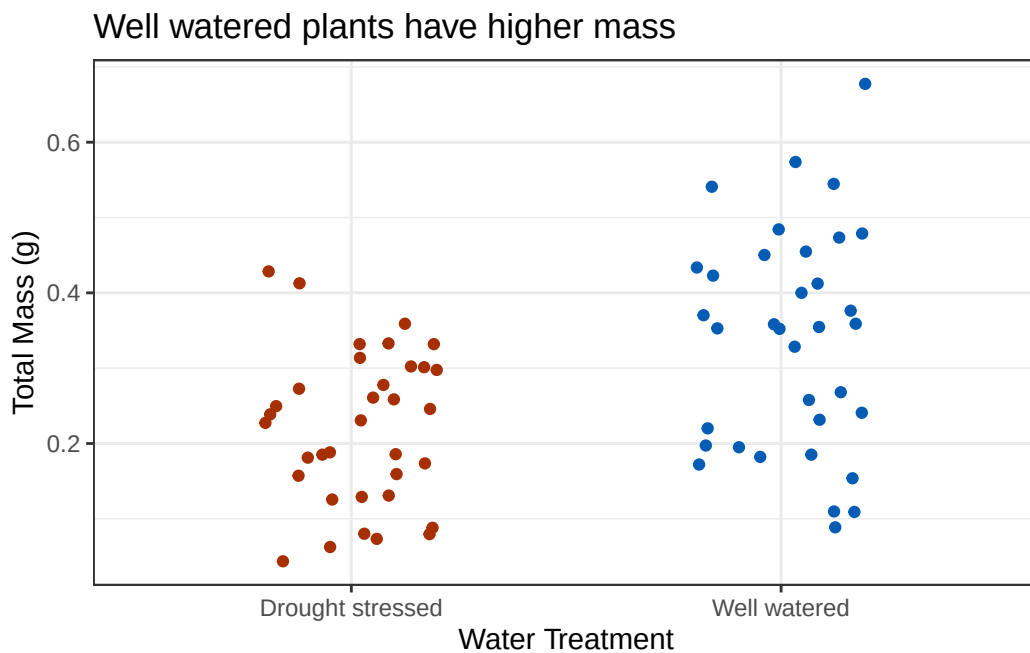
Add a title describing the differences in (mean or median) plant mass between water treatments.

```
# create base layer
# relate water treatment to total mass
ggplot(
  data = drought_exp_clean,
  mapping = aes(
```

```

    x = water_treatment,
    y = total_g,
    color = water_treatment
  )
) +
# create jitterplot, remove vertical jitter and narrow horizontal jitter
geom_jitter(height = 0,
            width = 0.2) +
# change colors using colors set earlier
scale_color_manual(
  values = c("Drought stressed" = ds_col,
            "Well watered" = ww_col)
) +
# rename axes, create title
labs(
  x = "Water Treatment",
  y = "Total Mass (g)",
  title = "Well watered plants have higher mass"
) +
# remove legend
theme(legend.position = "none")

```



What are the differences in total mass between species?

Make a jitterplot to visualize the differences in plant mass between species.

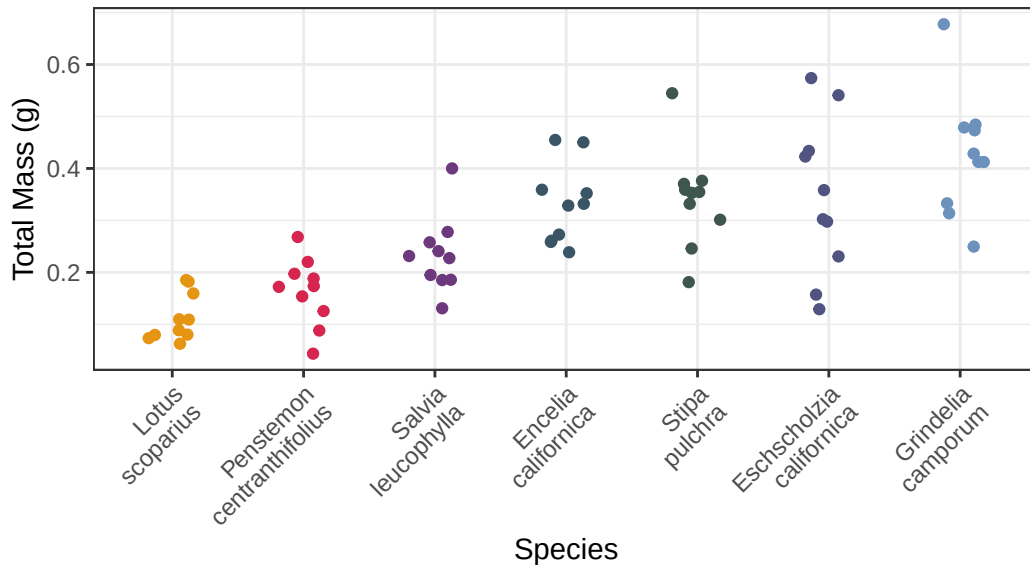
Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between species.

Bonus: order the x-axis by decreasing mean plant mass (no, I like the order it's in and I don't want to mess up the reference species)

```
# create base layer
# relate water treatment to total mass
ggplot(
  data = drought_exp_clean,
  mapping = aes(
    x = species_name,
    y = total_g,
    color = species_name
  )
) +
  # create jitterplot, remove vertical jitter and narrow horizontal jitter
  geom_jitter(height = 0,
              width = 0.2) +
  # rename axes, create title
  labs(
    x = "Species",
    y = "Total Mass (g)",
    title = "Grindelia has the highest mass and Lotus has \nthe lowest"
  ) +
  # remove legend
  # hjust ensures the text does not bleed into the chart
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1)) +
  # use plant colors set before
  scale_color_manual(values = plant_colors) +
  # wrap text to prevent long text strings
  scale_x_discrete(labels = label_wrap(10))
```


Grindelia has the highest mass and Lotus has the lowest



4. Fitting models

8 models total:

Model number	SLA	Water treatment	Species	Predictor list
0				no predictors (null model)
1	X	X	X	all predictors (full model)
2	X	X		SLA and water treatment
3	X		X	SLA and species
4		X	X	water treatment and species
5	X			SLA
6		X		water treatment
7			X	species

Model fitting

```
# model 0: null model
model0 <- lm(
  total_g ~ 1, # formula no predictors
```

```

    data = drought_exp_clean # use cleaned data
)

# model 1: all predictors
model1 <- lm(
  total_g ~ sla + water_treatment + species_name, # formula all predictors
  data = drought_exp_clean # use cleaned data
)

# model 2: SLA and water treatment
model2 <- lm(
  total_g ~ sla + water_treatment, # formula sla + water
  data = drought_exp_clean # use cleaned data
)

# model 3: SLA and species
model3 <- lm(
  total_g ~ sla + species_name, # formula sla + species
  data = drought_exp_clean # use cleaned data
)

# model 4: water treatment and species
model4 <- lm(
  total_g ~ water_treatment + species_name, # formula water + species
  data = drought_exp_clean # use cleaned data
)

# model 5: SLA
model5 <- lm(
  total_g ~ sla, # formula sla
  data = drought_exp_clean # use cleaned data
)

# model 6: water treatment
model6 <- lm(
  total_g ~ water_treatment, # formula water
  data = drought_exp_clean # use cleaned data
)

# model 7: species
model7 <- lm(
  total_g ~ species_name, # formula species

```

```
data = drought_exp_clean # use cleaned data
)
```

Dredge does this automatically, but avoid this because it causes you to avoid thinking about your data.

Model selection

Insert code to use the `AICc()` function to calculate the AIC for each model.

Arrange by decreasing AIC.

```
#|label: calc-AIC

# put all models into calculating AICc
AICc(
  model0,
  model1,
  model2,
  model3,
  model4,
  model5,
  model6,
  model7
) |>
# sorts the AICc column by the values in that column
arrange(AICc)
```

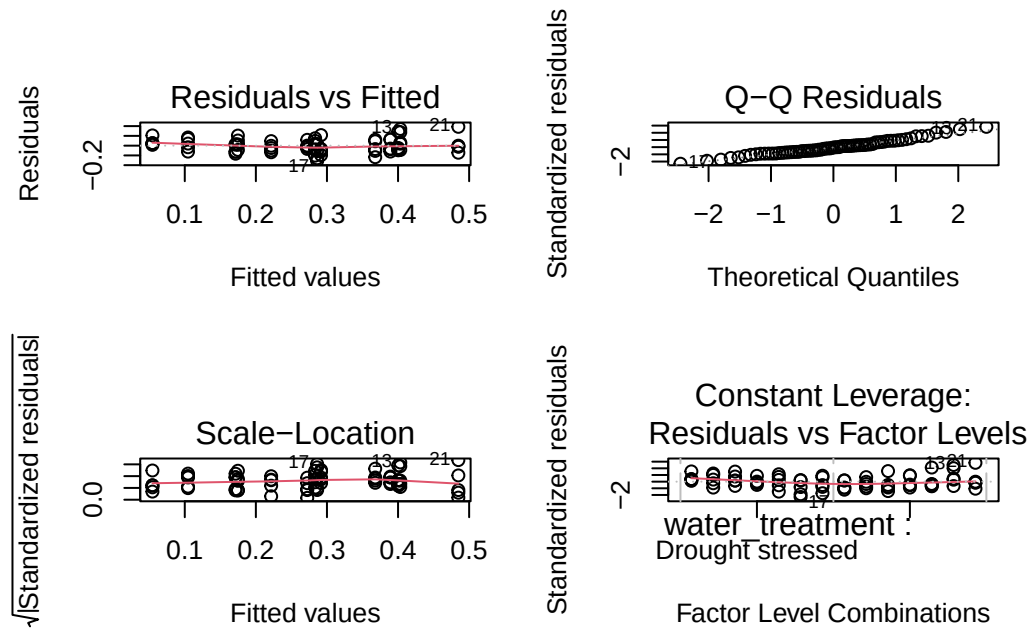
	df	AICc
model4	9	-156.19595
model11	10	-153.75361
model3	9	-124.07569
model7	8	-120.30191
model2	4	-95.82521
model5	3	-88.85180
model6	3	-86.76661
model0	2	-74.98036

```
# this shows that model 4 is the best model (best by >2)
# this is water treatment and species
```

Model Diagnostics

Check for best model

```
# display in 2x2 grid
par(mfrow = c(2, 2))
# display plots
plot(model4)
```



Clumps come from the fact it's categorical, not continuous. There's no vertical patterns.

Residuals are normally distributed and homoscedastic, and there's no outliers.

Insert code to display the summary for the best model.

```
# display summary for model 4
summary(model4)
```

Call:

```
lm(formula = total_g ~ water_treatment + species_name, data = drought_exp_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.157087	-0.046953	-0.003733	0.041244	0.192657

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.05455	0.02451	2.225	0.02973 *
water_treatmentWell watered	0.11695	0.01733	6.746	5.90e-
09 ***				
species_namePenstemon centranthifolius	0.05003	0.03243	1.543	0.12799
species_nameSalvia leucophylla	0.12020	0.03243	3.706	0.00045 ***
species_nameEncelia californica	0.21774	0.03243	6.714	6.70e-
09 ***				
species_nameStipa pulchra	0.22881	0.03243	7.055	1.72e-
09 ***				
species_nameEschscholzia californica	0.23164	0.03243	7.143	1.22e-
09 ***				
species_nameGrindelia camporum	0.31335	0.03243	9.662	5.53e-
14 ***				

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07252 on 62 degrees of freedom

Multiple R-squared: 0.7535, Adjusted R-squared: 0.7257

F-statistic: 27.08 on 7 and 62 DF, p-value: < 2.2e-16

Intercept: model prediction for plant mass for *Lotus scoparius* in drought stressed treatment (these are the two reference levels)

The other treatments/species predict how much higher plant mass will be if that category is present.

Ex: The well watered treatment has an estimate of 0.0546 with a standard error of 0.0245. This is the difference in *Lotus scoparius* mass between the drought stressed and well watered treatment

Penstemon centranthifolius has one of 0.05 with a standard error of 0.0324, note that $p > 0.05$, so this may be 0.

Stop and think: what does this model mean?

What is the best model?

The best model includes water treatment and species as predictors

How much variation in the response (total mass, in grams) does this model explain?

Adjusted R-squared is 0.726, so it's 72.6% of the variation

How do we interpret the effects of the predictors on the response variable (again, total mass in grams)?

Water treatment and species influence plant mass

4. Model predictions

Insert code to generate model predictions for the best model.

Rename the `x` column and `group` column to match the original data frame (`drought_exp_clean`).

Look at the object before plotting.

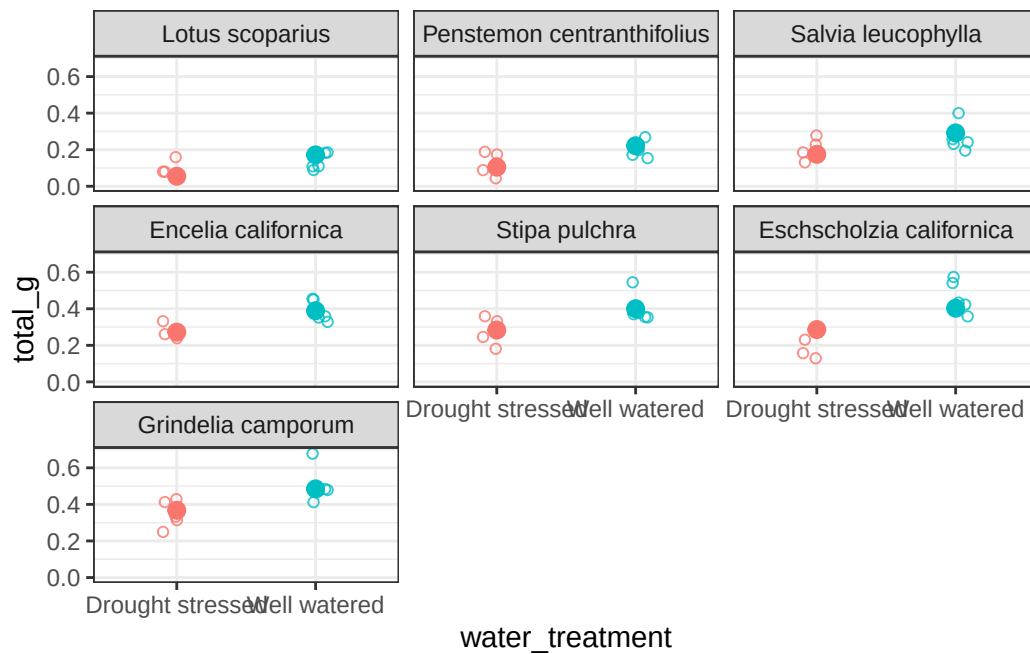
```
preds <- ggpredict(  
  model4,  
  terms = c("water_treatment", "species_name")  
) |>  
  rename(water_treatment = x,  
         species_name = group)
```

5. Final figure

Create a clean, publication ready figure depicting predictions for the best model and the underlying data.

```
ggplot(data = drought_exp_clean,  
       mapping = aes(  
         x = water_treatment,  
         y = total_g,  
         color = water_treatment  
       )) +  
  geom_jitter(  
    height = 0,  
    width = 0.1,  
    shape = 21,  
    alpha = 0.8  
  ) +
```

```
geom_pointrange(data = preds,
  mapping = aes(
    x = water_treatment,
    y = predicted,
    ymin = conf.low,
    ymax = conf.high
  )) +
facet_wrap(~ species_name) +
theme(legend.position = "none")
```



6. Writing

Figure caption

Basic components of test

predictors

response

test

distribution

parameters (degrees of freedom)

R^2

p-value

significance level

reference level(s)

Written communication

Methods

We used linear models to determine the predictive relationship between specific leaf area (cm^2/g), treatment, and species on plant mass (g). We evaluated model assumptions of normally distributed errors using QQ plots of model residuals. We evaluated assumptions of constant variance of errors using Residuals vs Fitted and Scale-Location plots. We used Akaike's Information Criterion (AIC) to select the simplest model that explained the most variation in plant mass.

Results

We found that the simplest model describing the greatest variation in plant mass included treatment and species as predictors (linear model: $F(2, 67) = 14.57$, $p < 0.001$, $\alpha = 0.05$).